

# DW\_fp\_div

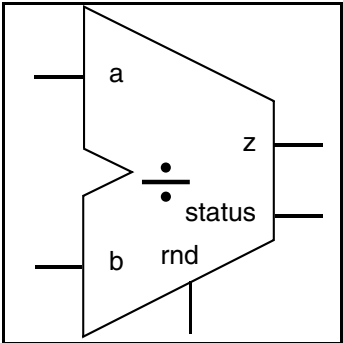
## Floating-Point Divider

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

### Features and Benefits

- The precision format is parameterizable for either IEEE single, double precision, or a user-defined custom format
- Hardware for denormal numbers of IEEE 754 standard is provided
- Configurable to be fully compliant with the IEEE Std 754-1985 standard
- Configurable for NaN representation compatible with the IEEE Std 754-2008 standard (controlled by the *ieee\_compliance* parameter)
- Faithful rounding with 1 ulp error is supported by parameter
- DesignWare datapath generator is employed for better timing and area

### Revision History



### Description

DW\_fp\_div is a floating-point divider that divides two floating-point operands: a by b to produce a floating-point quotient, z.

Component pins are described in [Table 1-1](#) and configuration parameters are described in [Table 1-2](#).

Table 1-1 Pin Description

| Pin Name | Width                                        | Direction | Function                                                                                                                                                                                |
|----------|----------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a        | <i>exp_width</i> + <i>sig_width</i> + 1 bits | Input     | Dividend                                                                                                                                                                                |
| b        | <i>exp_width</i> + <i>sig_width</i> + 1 bits | Input     | Divisor                                                                                                                                                                                 |
| rnd      | 3 bits                                       | Input     | Rounding mode; supports all rounding modes described in the <a href="#">Datapath Floating-Point Overview</a> ;<br>The <i>rnd</i> port takes effect only when <i>faithful_round</i> = 0. |
| z        | <i>exp_width</i> + <i>sig_width</i> + 1 bits | Output    | Quotient of A/B                                                                                                                                                                         |

Table 1-1 Pin Description (Continued)

| Pin Name | Width  | Direction | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|--------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| status   | 8 bits | Output    | <ul style="list-style-type: none"><li>Status flags corresponding to <i>z</i>; for details, see <a href="#">STATUS Flags</a> in the <i>Datapath Floating-Point Overview</i></li><li>status[6]: Underflow before rounding (UBR) status flag<br/>When enabled by the <i>en_ubr_flag</i> parameter, this flag indicates when the absolute value of a non-zero result, computed as though both the exponent range and the precision of the significand were unbounded, would lie strictly between <math>\pm\text{MinNorm}</math> (minimum normalized value representable in the FP format defined by <i>sig_width</i> and <i>exp_width</i>).</li><li>status[7]: Indicates divide-by-zero operation</li></ul> |

Table 1-2 Parameter Description

| Parameter       | Values                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sig_width       | 2 to 253 bits<br>Default: 23 | Word length of fraction field of floating-point numbers <i>a</i> , <i>b</i> , and <i>z</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| exp_width       | 3 to 31 bits<br>Default: 8   | Word length of biased exponent of floating-point numbers <i>a</i> , <i>b</i> , and <i>z</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ieee_compliance | 0, 1, or 3<br>Default: 0     | <p>Level of support for the IEEE Std 754 standards:</p> <ul style="list-style-type: none"><li>0: No support for NaNs and denormals; NaNs are considered infinities and denormals are considered zeros</li><li>1: Fully compliant with the IEEE Std 754-1985 standard, including support for NaNs and denormals</li><li>2: Reserved</li><li>3: Fully compliant with the IEEE Std 754-1985 standard plus NaN representation that matches the IEEE Std 754-2008 standard<sup>a</sup></li></ul> <p>For details, see <a href="#">Compatibility with IEEE Std 754 Standards</a> in the <i>Datapath Floating-Point Overview</i></p>                                                                                                                                                                                                                                                    |
| faithful_round  | 0 or 1<br>Default: 0         | <p>Choose either a specific rounding mode (set by <i>rnd</i>) or a general rounding mode that allows maximum 1 ulp error</p> <ul style="list-style-type: none"><li>0: Rounding mode is specific, as set by the <i>rnd</i> port; this choice increases the size of the resulting implementation.</li><li>1: Rounding mode is general and, for <i>sig_width</i> <math>\leq</math> 23, allows a maximum of 1 ulp error<sup>b</sup>; this choice decreases the size of the resulting implementation.</li></ul> <p>When <i>faithful_round</i> = 1, note the following:</p> <ul style="list-style-type: none"><li>The inexact status flag in the output is not meaningful.</li><li>The UBR status flag (<i>status</i>[6]) is always 0 (disabled).</li><li>The other status flags will match one of the possible outputs for the calculation when <i>faithful_round</i> = 0.</li></ul> |

**Table 1-2 Parameter Description (Continued)**

| Parameter   | Values               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| en_ubr_flag | 0 or 1<br>Default: 0 | Controls when the flag in <code>status[6]</code> is enabled <ul style="list-style-type: none"> <li>0: The UBR status flag <code>status[6]</code> is disabled (the value of this flag is always 0).</li> <li>1: When this parameter is 1 and <code>faithful_round = 0</code>, <code>status[6]</code> carries the value of the UBR flag. If <code>faithful_round = 1</code>, even when <code>en_ubr_flag = 1</code>, the UBR flag in <code>status[6]</code> bit is always 0.</li> </ul> |

- Propagating payload information to the output during the NaN process, which is an optional feature specified in the IEEE Std 754-2008 standard, is not supported.
- When `faithful_round = 1` and `sig_width > 23`, the result can exceed 1 ulp for some corner cases. Configurations tested for this parameter range do not show errors larger than 2 ulps.

**Table 1-3 Synthesis Implementations<sup>a</sup>**

| Implementation Name | Function                                                                                                                                                                                         | License Feature Required |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| rtl                 | Synthesis model (Digit-recurrence Method)                                                                                                                                                        | DesignWare               |
| str                 | Synthesis model (Newton-Raphson Method)<br>NOTE: This implementation is disabled when <code>sig_width ≤ 10</code> . In this scenario, use the rtl implementation because it provides better QoR. | DesignWare               |

- During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

**Table 1-4 Simulation Models**

| Model                         | Function                             |
|-------------------------------|--------------------------------------|
| DW02.DW_FP_DIV_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw02/src/DW_fp_div_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW_fp_div.v        | Verilog simulation model source code |

## Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW\_SUPPRESS\_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If an invalid rounding mode has been detected on `rnd`, the following message is displayed:

```
WARNING: <instance_path>:  
        at time = <timestamp>: Illegal rounding mode.
```

To suppress this message, use the DW\_SUPPRESS\_WARN macro explained earlier.

## Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

## HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp.all;
-- If using numeric types from std_logic_arith package,
-- comment the preceding line and uncomment the following line:
-- use DWARE.DW_Foundation_comp_arith.all;

entity DW_fp_div_inst is

    generic (
        inst_sig_width      : POSITIVE := 23;
        inst_exp_width      : POSITIVE := 8;
        inst_ieee_compliance : INTEGER := 0;
        inst_faithful_round  : INTEGER := 0;
        inst_en_ubr_flag    : INTEGER := 0
    );
    port (
        inst_a      : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        inst_b      : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        inst_rnd     : in std_logic_vector(2 downto 0);
        z_inst      : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        status_inst : out std_logic_vector(7 downto 0)
    );

end DW_fp_div_inst;

architecture inst of DW_fp_div_inst is

begin

    -- Instance of DW_fp_div
    U1 : DW_fp_div
    generic map (
        sig_width => inst_sig_width,
        exp_width => inst_exp_width,
        ieee_compliance => inst_ieee_compliance,
        faithful_round => inst_faithful_round,
        en_ubr_flag => inst_en_ubr_flag
    )
    port map (
        a => inst_a,
        b => inst_b,
        rnd => inst_rnd,
        z => z_inst,
        status => status_inst
    );

```

```
end inst;
```

```
-- pragma translate_off  
configuration DW_fp_div_inst_cfg_inst of DW_fp_div_inst is  
  for inst  
    end for;  
end DW_fp_div_inst_cfg_inst;  
-- pragma translate_on
```

## HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_div_inst( inst_a, inst_b, inst_rnd, z_inst, status_inst );

parameter sig_width = 23;
parameter exp_width = 8;
parameter ieee_compliance = 0;
parameter faithful_round = 0;
parameter en_ubr_flag = 0;

input [sig_width+exp_width : 0] inst_a;
input [sig_width+exp_width : 0] inst_b;
input [2 : 0] inst_rnd;
output [sig_width+exp_width : 0] z_inst;
output [7 : 0] status_inst;

// Instance of DW_fp_div
DW_fp_div #(sig_width, exp_width, ieee_compliance, faithful_round, en_ubr_flag) U1
( .a(inst_a), .b(inst_b), .rnd(inst_rnd), .z(z_inst), .status(status_inst)
);

endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | ■ Updated date and version; no technical changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| December 2023  | DWBB_202312.0 | ■ Updated date and version; no technical changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| April 2022     | DWBB_202203.1 | ■ Added the description of <code>status[6]</code> to <a href="#">Table 1-1</a> on page 1<br>■ Added the <code>en_ubr_flag</code> parameter to <a href="#">Table 1-2</a> on page 2 and to the examples on <a href="#">page 5</a> and <a href="#">page 7</a>                                                                                                                                                                                                                                                                                                                                                                              |
| January 2022   | DWBB_202106.5 | ■ Adjusted description of the <code>str</code> implementation in <a href="#">Table 1-3</a> on page 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| October 2020   | DWBB_202009.1 | ■ For enhanced NaN compatibility with the IEEE Std 754 standards, added a new value for <code>ieee_compliance</code> in <a href="#">Table 1-2</a> on page 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| July 2020      | DWBB_201912.5 | ■ Adjusted the description of the <code>ieee_compliance</code> parameter in <a href="#">Table 1-2</a> on page 2<br>■ Added “ <a href="#">Suppressing Warning Messages During Verilog Simulation</a> ” on page 4<br>■ Added the <code>faithful_round</code> parameter to the examples on <a href="#">page 5</a> and <a href="#">page 7</a>                                                                                                                                                                                                                                                                                               |
| April 2020     | DWBB_201912.3 | ■ Updated the value range for the <code>sig_width</code> parameter in <a href="#">Table 1-2</a> on page 2<br>■ Updated the description of <code>rnd</code> in <a href="#">Table 1-1</a> on page 1 and <code>faithful_round</code> in <a href="#">Table 1-2</a> on page 2<br>■ For STAR 3124396, added a footnote to <a href="#">Table 1-2</a> on page 2 to update the error range when <code>faithful_round = 1</code> and <code>sig_width &gt; 23</code> . This update is based on a limitation found during the investigation of STAR 3124396. Also, <a href="#">Table 1-5</a> , “Error Ranges ( $\epsilon = z - a/b$ )” was removed. |
| July 2017      | DWBB_201612.5 | ■ For STAR 9001189734, added Note to description of <code>str</code> implementation in <a href="#">Table 1-3</a> on page 3<br>■ Added this Revision History table                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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